Market Entry Strategy for a Coffee Company

Market segmentation and regional prioritisation using Google business listings (Ukraine case study)

1. Executive summary

- **Question:** How is the existing coffee shop market structured, and where should a new entrant open first?
- **Segments:** Four segments describe the market: established full-service leaders (33 shops), a mainstream dine-in/takeout core (92), newcomer grab-and-go (33) and newcomer small-format/niche (38).
- **Rating is not a differentiator:** 52% of shops are rated 4.7 or higher, and 95% of priced shops are mid-range ($$). Entry should compete on service consistency and format, not price.
- **Format matters:** Shops offering both dine-in and takeout have a median of 485 reviews versus 64 for those offering neither. Only 17% of shops offer delivery.
- **Prediction:** Review volume can be predicted for unseen shops (cross-validated R-squared 0.65, typical error about 2x), but neural networks did **not** beat simpler models on 196 shops, and much of the signal is listing completeness rather than an actionable lever.
- **Recommendation:** Enter Odessa first (youngest market), then Dnipro (lowest average rating, suggesting a service gap), with Kyiv as a later scale play. Poltava, shortlisted in an earlier draft, is not supported by the data.

2. Data and cleaning

The dataset contains 200 coffee shop listings (20 per region across 10 regions) with rating, review count, price band and dine-in, takeout and delivery flags. Cleaning steps left 196 shops:

- Dropped 2 listings with no rating or reviews, 1 appliance-repair business, and 1 coffee manufacturer with 17,937 reviews (an extreme outlier that would distort clustering).
- Dine-in and takeout flags are only shown on Google when true, so missing values were treated as *not offered*. Delivery has both true and false values, so missing is treated as unknown/not offered.
- Reviews are heavily skewed, so a log transform was applied before clustering.

Important limits. Exactly 20 shops were sampled per region, so shop counts say nothing about market size. Review count measures how established a shop is, not its revenue. The data appears to pre-date 2022 (it includes Mariupol), so it must be refreshed before any real investment decision, and security conditions in frontline cities should be checked independently.

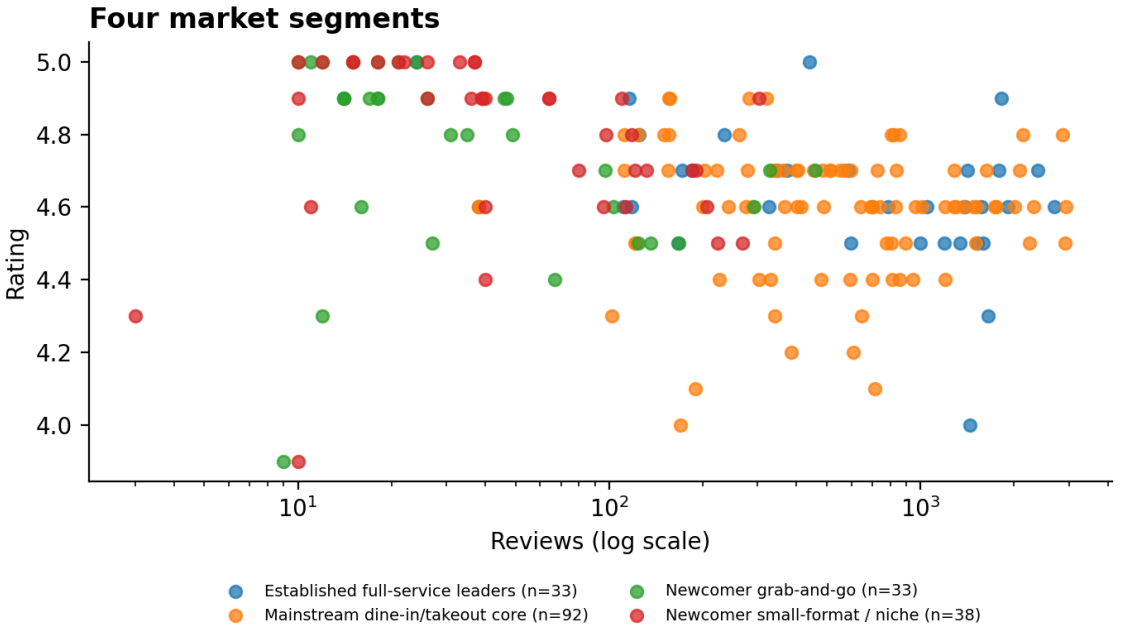
3. Method

Shops were grouped with K-means clustering on five standardised features: rating, log(reviews), dine-in, takeout and delivery. Silhouette scores are nearly flat for 3 to 5 clusters (about 0.40 for each), so the data does not strongly prefer one number of segments. Four was chosen because it gives interpretable, actionable groups, and the result should be presented as a useful lens rather than a hard fact. Regions were compared on median reviews (robust to outliers), the share of shops with fewer than 100 reviews, average rating and delivery availability.

4. Findings

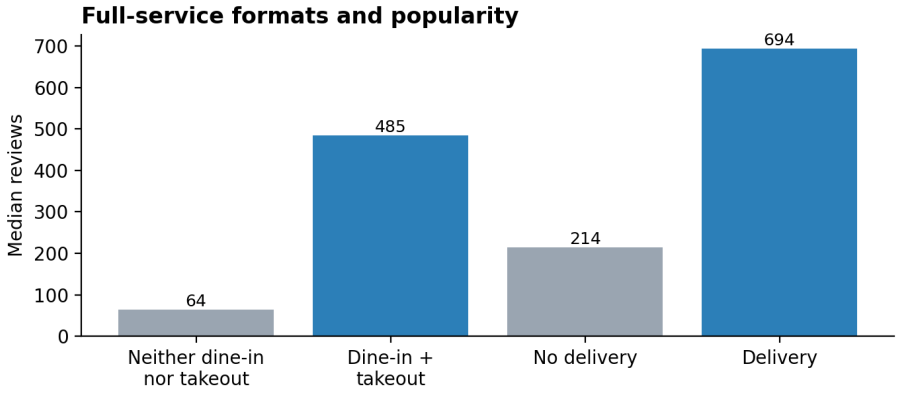
4.1 Market segments

Segment	Shops	Median reviews	Avg rating	Dine-in	Takeout	Delivery
Established full-service leaders	33	789	4.63	85%	70%	100%
Mainstream dine-in/takeout core	92	560	4.59	100%	95%	0%
Newcomer small-format / niche	38	40	4.78	24%	0%	3%
Newcomer grab-and-go	33	24	4.75	30%	100%	0%



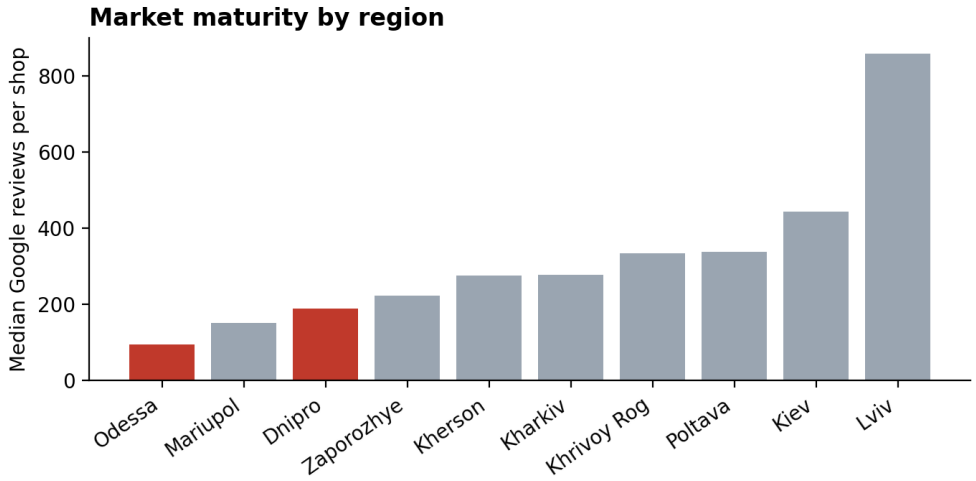
The mainstream core is the largest and most contested group, and it has the lowest average rating (4.59) despite lots of reviews. Newcomers rate higher (4.75 to 4.78) because young listings have fewer, friendlier reviews. Across all shops, rating and review count are negatively correlated (Spearman -0.44), so a high rating alone says little about quality.

4.2 Format and popularity



Full-service formats go with higher review counts. This is a correlation, since larger, older shops tend to add services, but because delivery is still rare it is a visible gap for a new entrant.

4.3 Regional maturity



Region	Median reviews	Shops with <100 reviews	Shops with >1,000 reviews	Avg rating	Delivery
Odessa	96	50%	15%	4.78	20%
Mariupol	151	35%	0%	4.70	20%
Dnipro	189	32%	21%	4.55	21%
Zaporozhye	223	26%	11%	4.65	16%
Kherson	275	37%	11%	4.64	16%
Kharkiv	278	20%	15%	4.58	10%
Khrivoy Rog	334	20%	10%	4.60	25%
Poltava	338	30%	20%	4.65	20%
Kiev	444	30%	35%	4.78	15%
Lviv	858	11%	47%	4.67	11%

- **Odessa:** lowest median reviews (96) and half of shops under 100 reviews, so incumbents are young. The 4.78 average rating means the quality bar is high.
- **Dnipro:** lowest average rating (4.55) with nearly 9 in 10 shops offering dine-in, which suggests room to win on service quality.
- **Kyiv and Lviv:** the most mature markets (35% and 47% of shops above 1,000 reviews). Hardest to break into, but Kyiv is the likely scale opportunity later.
- **Poltava:** median reviews of 338 rank it 8th of 10 and 20% of its shops exceed 1,000 reviews, so it is not an early-entry market on this evidence.

5. Recommendation

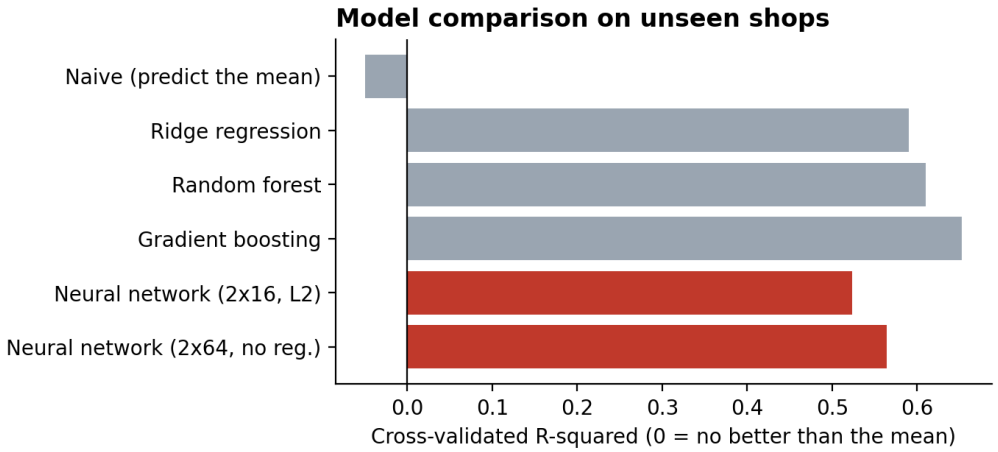
- **Concept:** mid-price, full format (dine-in, takeout and delivery) in a cafe or coffee-shop layout.
- **Sequence:** Odessa first, Dnipro second, then Kyiv once the playbook is proven.
- **First 90 days:** a review-generation programme (for example, a small discount for an honest review), because review volume is how listings gain visibility and trust.
- **Positioning:** compete on consistent service and format breadth, since price and rating do not separate competitors.

6. Predictive modelling: what explains shop popularity?

6.1 Question and approach

Review count is the best available proxy for how established a shop is and how much demand it captures. The question is whether it can be predicted from observable attributes, and which attributes matter. **Target:** log10(reviews). **Features:** rating, dine-in, takeout, delivery, price listing and band, place type and region. **Evaluation:** 5-fold cross-validation repeated 5 times, scored only on shops the model had not seen. R-squared is the share of variation explained, where 0 means no better than guessing the average.

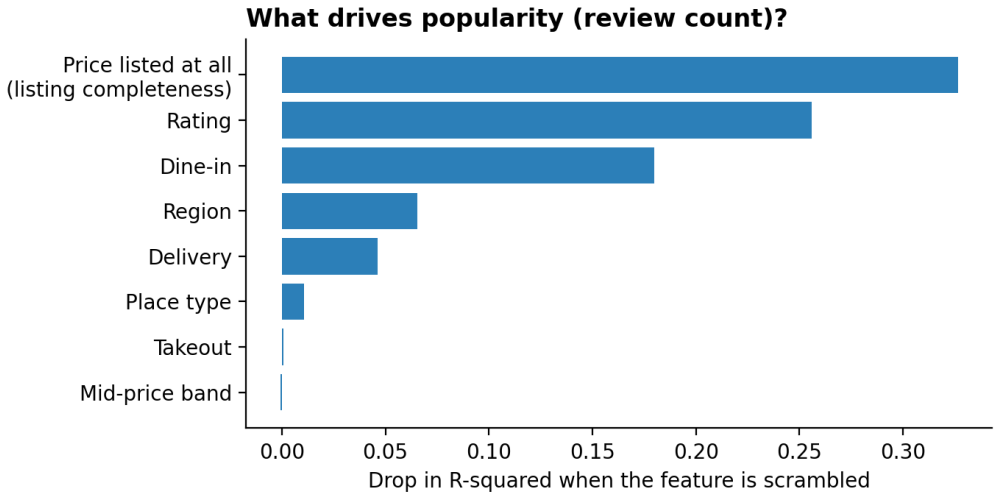
Model	R-squared (CV)	Typical error
Naive baseline (predict the mean)	-0.05 (+/- 0.08)	about 4.0x
Ridge regression (linear)	0.59 (+/- 0.06)	about 2.2x
Random forest	0.61 (+/- 0.08)	about 2.1x
Gradient boosting	0.65 (+/- 0.08)	about 2.0x
Neural network, 2x16, regularised	0.52 (+/- 0.08)	about 2.3x
Neural network, 2x64, lightly regularised	0.56 (+/- 0.09)	about 2.3x



6.2 Neural networks were tested, but simpler models win

Two feed-forward neural networks (two hidden layers, trained by backpropagation) scored 0.56 and 0.52, below gradient boosting at 0.65 and even below plain linear regression at 0.59. With about 160 training shops per fold, a neural network has far more parameters than the data can support, and stronger regularisation did not rescue it. **Recommendation:** use gradient boosting (or the interpretable ridge model) for this dataset. Deep learning becomes worthwhile with thousands of listings, or when analysing review text with language models.

6.3 What drives popularity, and what does not



Feature set	R-squared (CV)
All features	0.65
Without price-listed flag	0.64
Without price-listed, dine-in and takeout flags	0.36
Rating and region only	0.32

The top signals are whether a price is listed at all, the rating, and the dine-in flag. The first and third are largely **listing completeness**: shops with a price listed have a median of 587 reviews versus 38 for those without, and shops with a dine-in flag have 479 versus 40. These flags are symptoms of an established, well-maintained listing, not levers a new entrant can pull to gain popularity. When they are removed, predictive power falls from 0.65 to 0.36. The practical takeaway is modest but sensible: fill in a complete Google profile from day one. Mid-price band and takeout add almost nothing once other features are known.

Holding other features constant (ridge model), a shop with a dine-in flag has about 1.6x the reviews of one without, a shop with delivery about 1.3x, and a one standard deviation (about 0.22 star) higher rating goes with 0.79x the reviews, because newer shops rate higher. These are associations, not proven effects.

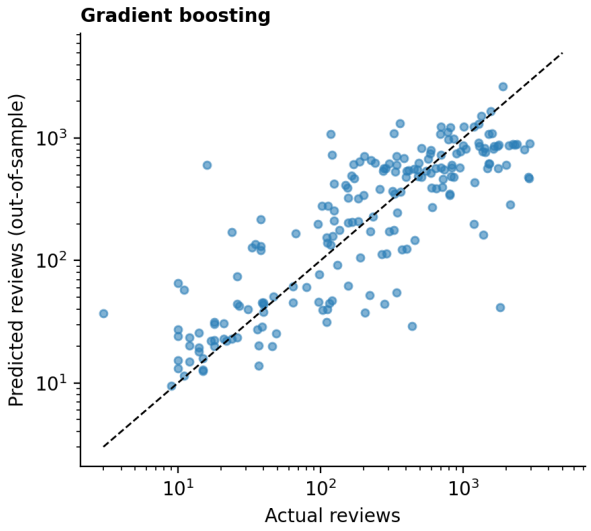
6.4 Regional effect after adjusting for format and rating

Region	Review multiplier	Region	Review multiplier
Kiev	1.89x	Kherson	0.94x
Lviv	1.56x	Kharkiv	0.91x
Zaporozhye	1.04x	Odessa	0.88x
Khrivoy Rog	1.00x	Dnipro	0.75x
Poltava	0.99x	Mariupol	0.59x

Multipliers show review volume relative to the average region. Kyiv (1.9x) and Lviv (1.6x) are the most review-intensive, while Mariupol (0.6x), Dnipro (0.75x) and Odessa (0.88x) sit below average. This fits the earlier ranking, but it has two possible readings: *immature competition* (an opportunity) or *smaller demand* (a risk). Reviews reflect customer volume multiplied by shop age, so this dataset cannot separate the two. Population, income and footfall data are needed before treating Odessa or Dnipro as proven.

6.5 Implications for the client

- **Choice of tool:** deploy a gradient boosting benchmark, not deep learning, at the current data scale.
- **Use as a benchmark:** predicted versus actual reviews can flag competitors that outperform their profile. These are the shops worth studying for what they do differently.
- **Do not over-read the drivers:** the strongest predictors are proxies for maturity, so they support the 'established versus newcomer' segmentation but not a causal playbook.
- **Cheap action:** complete every Google profile field at launch (price, hours, dine-in, takeout, delivery).



Out-of-sample predictions follow the actual review counts but with wide scatter (typical error about 2x), so the model suits benchmarking and ranking, not precise forecasts.

7. Limitations and next steps

- Sample of 196 shops, collected with fixed quotas per region, cannot size the market. Add population, income, rent and foot-traffic data.
- Add chain competitors, unit economics and a break-even estimate to turn the attractiveness ranking into a financial case.
- Test sensitivity of the segment count (3, 4, 5) and of the outlier rule.
- Read a sample of review text to find specific service gaps (waiting time, staff, seating).
- Collect shop opening dates and review timestamps. Reviews per month would separate shop age from true demand, and is the single most valuable missing variable.
- Once there are thousands of listings, or review text is available, revisit deep learning (for example, language-model sentiment or topic analysis).

Data attribution: The analysis uses the coffee_shops.csv dataset of Google business listings supplied with the original exercise. Credit the original source and author in your repository README before publishing.